GeM💎NR: Geometry-Aware Multi-View Editing for Nonrigid Scene Changes

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`https://gem-nr.github.io`

Abstract. Recent developments in multi-view image editing with generative models have brought us a step closer toward general 3D content generation and customization. Most existing works focus on *rigid* or *appearance-only* edits by utilizing the geometry of the unedited scene. This naturally limits these methods to edits that preserve the underlying scene structure. Current nonrigid approaches are limited to object removal and insertion, reflecting the data they are trained on. General nonrigid edits, *i.e.*, edits that substantially and arbitrarily change the scene geometry, remain challenging for existing methods. We propose GeM-NR, a fast and flexible training-free approach for *general* multi-view consistent image editing, including edits that drastically change the geometry and appearance of the scene. Given an anchor image edited with a chosen 2D editor and a query unedited image, GeM-NR edits the query image consistently with the anchor edit. The method incorporates multiple stages: (i) depth map estimation, where we propose a strategy to maximize the alignment between the 3D point clouds of the edited and unedited scenes, (ii) projection onto a query viewpoint, and (iii) refinement of the obtained image conditioned on the unedited query. We demonstrate the ability of our method to handle edits with significant changes in geometry and appearance, something that existing methods struggle with. We perform an extensive evaluation showing that GeM-NR improves consistency for a wide variety of edit tasks, including generating 3D representations of the edited scene. Both quantitative and qualitative results indicate the state-of-the-art performance of our method in terms of edit quality as well as geometric and photometric consistency across multiple views.

1 Introduction

Consistent multi-view image editing is a core capability for 3D editing, with numerous possible applications. A major challenge is in handling nonrigid edits, which drastically change the geometry of the scene. One reason for difficulties with these types of edits is that the geometry of the original unedited scene

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Fig. 1: The edited images produced by GeM-NR. We handle various types of edits including significant changes in scene geometry. The edits are both photometrically and geometrically consistent across the views.

no longer holds for the edited scene. Another reason is that the more drastic edits often lead to significant inconsistencies across different views, making the task of enforcing consistency more difficult. One approach for solving this is to train a method using paired multi-view images with such edits [37]. However, limitations in the availability of paired data restrict these methods to specific types of edits, such as object insertion and removal.

One solution to the issue with significant inconsistencies in the edits is to warp an edited image into a target view and use that as conditioning when editing an image at that view, providing clear guidance on how that view should look to be consistent with the previous edit. Existing works [17, 31] use depth maps acquired from a 3D representation of the original scene, limiting the ability to handle nonrigid edits. In contrast, we propose estimating the scene geometry from an edited view of the scene using a monocular depth estimator [33], since these depths will then be valid also for nonrigid edits. We formulate this problem as the task of performing consistent editing of a number of images of the same scene, given an initial edited image. This problem could be solved by performing single-image novel view synthesis from the initial edited image by rendering new views at the poses of the given views. One approach is to estimate geometry from the initial image and use this to warp the initial image into the target views. The problem with doing this in the setting of multi-view editing is that this does not take into account the information in the additional views of the scene. We instead propose a way to include this warping in the editing process, so that both the unedited views and the warping of the initial edit are taken into account.

Recent developments in editing methods allow for flexible and multi-reference editing [6,7,61], which could be used for multi-view editing. It is possible to condition these methods on several views and task these methods with performing multi-view consistent editing, but this is unstable and often leads to failed edits. Our contribution is that by conditioning these multi-reference methods with both an unedited image and the warping of an existing edit, it is possible to reliably perform multi-view consistent editing.

While we rely on foundation 3D models for reconstruction and multi-reference image generation models for editing, we discover novel ways of leveraging their power. In particular, our first finding is that the unedited and edited scenes together can be cast as one dynamic scene, and hence can be handled by a dynamic scene reconstruction model such as Depth Anything 3. Our second finding is that the unedited image together with its partially filled edited version is a better input for the multi-reference image generation model such as FLUX.2 than a fully edited image from another viewpoint. It has been shown that image generation models have limited ability in performing geometric tasks. However, if provided with the right geometry, they succeed at preserving it.

In summary, for an initial edit done by a preferred backbone editor, GeM-NR is able to perform multi-view consistent editing with respect to this initial edit. GeM-NR is flexible and can handle widely varied and drastic edits, including significant changes to geometry, as seen in Figure 1. It does not require any per-scene optimization and the runtime is only limited by the time it takes to perform monocular depth estimation and multi-reference editing, which for the methods we choose can be done in a few seconds per image. Our contributions can be summarized as follows:

- We propose a flexible pipeline for multi-view image editing based on reconstructing the geometry of the edited scene;
- We discover and combine novel ways of leveraging the power of foundation models, in particular dynamic reconstruction models and multi-reference image editing models, to maximize the editing quality *together* with multi-view consistency;
- We conduct extensive evaluation showing that GeM-NR can handle edits significantly changing geometry and appearance. We propose a more detailed evaluation pipeline integrating epipolar geometry evaluation, showing improved consistency across a variety of different editing tasks and image editing methods;
- Our multi-view pipeline is very fast, 2-4× faster than the state of the art. This allows us to generate edited 3D Gaussians in less than a minute for sparse scenes.

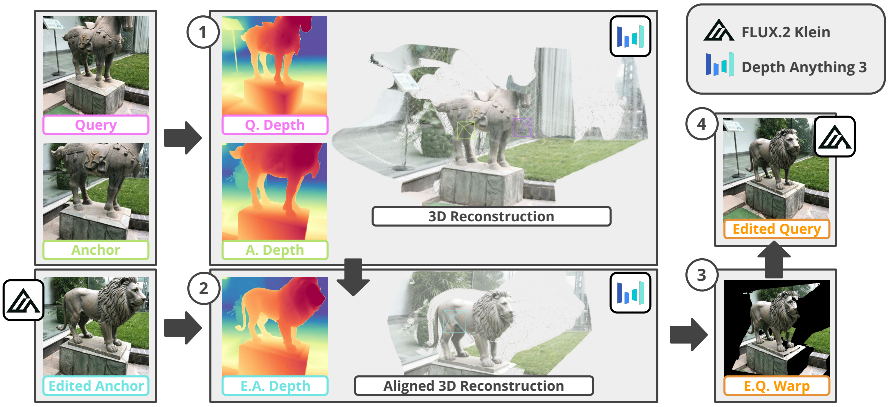
2 Related Work

Image editing using generative models. Diffusion models trained on vast amounts of data have led to significant advancements in both image generation [25,46,50,52] and editing [8,23,39,51]. This has enabled different types of editing tasks, including instruction-based editing [8,20,71], image inpainting [27,38,76], style transfer [11,15,24,73], and multi-reference editing [6,7,61,64]. Recent developments in flow matching [34,36] and multi-modal transformer architectures [19] have led to high-quality unified editing models [6,7,29,61] that can reliably perform significant and precise edits. Our work focuses on using these powerful 2D image editing methods to perform consistent multi-view editing.

Multi-view consistent image editing. Much of the research on multi-view consistent editing focuses on adapting 2D image editing models to edit an existing 3D representation, such as NeRFs or 3D Gaussians. Instruct-NeRF2NeRF [22] introduces iterative dataset update (IDU), which achieves consistent edits by iteratively editing views and updating the 3D representation. This strategy is further adopted in several works [14,17,40,56,60]. Other approaches leverage the 3D Gaussian representation of the unedited scene, e.g., by using its geometry to guide the editing process [28,31,62,75], or by rendering smooth sequences and using video diffusion priors to maintain 3D consistency [12,13,44]. A related technique is to iteratively update the 3D representation with edited images, thereby inheriting consistency from the underlying representation [12,13,30,31,60,62]. These approaches require dense views to obtain a high-quality 3D representation of the unedited scene. An alternative that avoids this requirement is to use correspondences between the unedited images to guide the editing such that corresponding points are edited consistently [2,4,54]. While directly encouraging multi-view consistency, it has limitations for edits that significantly change the geometry of the scene, since the correspondences from the unedited images do not hold after the edit. Another line of work directly edits a 3D representation or a set of multi-view images, but requires paired 3D data, limiting most methods to single objects and synthetic scenes [5,32,63,68]. Recent works try to extend this approach to real images, either by using a pipeline based on developments in 2D image editing and VLMs [16,65] to create paired multi-view edits for specific edit types [37] or by performing reinforcement-learning-based finetuning and leveraging the 3D foundation model VGGT [57] as a consistency reward [58]. In contrast, GeM-NR is training-free and does not require dense views: we estimate the geometry of the edited scene from the initial edit and use it to compute a warping that conditions the editing of the second view, propagating the edit consistently even for nonrigid scene changes.

Warping-based conditioning. One way to improve multi-view consistency is to warp an input view to a target viewpoint. This has been used for single-image novel view synthesis, where an input image is warped to a target pose, conditioning the generation for the corresponding view [35,42,43,53,55,69,70]. Several works also use warping to improve consistency in 3D editing, either by leveraging depth from the unedited 3D representation to blend edits across views [17,31], conditioning the editing on estimated depths [49,62], or warping attention feature maps [21]. These approaches are primarily designed for rigid edits that roughly preserve the scene geometry, since the depth is derived from the unedited scene. In contrast, we propose to utilize the geometry of the edited scene, making the

Fig. 2: Method overview. GeM-NR estimates the geometry of the edited scene while maintaining global alignment with the unedited scene, projects the resulting point cloud onto the target view, and uses the resulting warp to condition the editing of the next image.



approach applicable to nonrigid scene changes. Recent geometry estimation models [33, 47, 48, 66] can be leveraged to obtain the geometry of the edited scene from the initial edited image, and to warp the image into a target view. Our approach adapts this for nonrigid multi-view editing by combining the warped edit with the image to be edited to preserve multi-view details and consistency even when the edit changes scene geometry.

3 Method

3.1 Problem formulation

Given N source views $\{I^1_{src}, I^2_{src}, ..., I^N_{src}\}$ and a text description T of the desired edit, the goal is to edit the images, obtaining $\{I^1_{edited}, I^2_{edited}, ..., I^N_{edited}\}$, such that their appearance and geometry are consistent across views, while remaining faithful to the edit description T. One way to reformulate the problem is to first choose an anchor image A_{src} that is to be edited independently: $A_{edited} = f(A_{src}, T)$, where f is a preferred image editing model. The remaining images, which we refer to as query images and denote by $\{Q^i_{src}\}$, can then be edited conditioned on A_{edited} for consistent editing. Thus, we simplify the task to processing triplets consecutively: given $\{Q_{src}, A_{src}, A_{edited}\}$, the goal is to produce Q_{edited} that is consistent with A_{edited}. As of now, existing image editing models do not guarantee that independently edited images, that is, $Q_{edited} = f(Q_{src}, T)$, will retain such consistency.

3.2 Overview

Our editing pipeline should be able to handle significant changes in the scene. For appearance changes, since at least two unedited views are available, one

Fig. 3: Monocular vs. our joint depth estimation. In regions where the geometry is unchanged by the edit (zoomed-in), the depths of the unedited and edited scene from the same viewpoint should coincide. Monocular depths are not as accurately aligned w.r.t the unedited scene as our jointly estimated depths, which use all unedited and edited images.

could use the geometry of the unedited scene obtained with a 3D reconstruction pipeline. For challenging geometric changes, however, we cannot utilize the geometry acquired from the unedited images. Instead, we propose to use the geometry of the edited scene. Moreover, we need to understand how the geometry of the edited scene relates to that of the unedited scene. In other words, the two 3D representations need to be aligned. However, solving the alignment accurately and robustly is challenging, especially if the majority of the scene geometry is changed. We sidestep the alignment problem altogether and instead aim to jointly estimate the 3D representations of both the edited and unedited scene as realizations of one dynamic scene, which automatically results in a global alignment. This becomes possible with the recent advances in dynamic scene reconstruction [33]. The 3D representation of the globally aligned edited scene can now be used to render an image at a query viewpoint. Naturally, at this viewpoint, some regions need to be filled in as they were not seen from the anchor viewpoint. A source query image provides additional information about what should be depicted in these areas. A multi-reference image generation model [6, 7] can therefore be used to fulfill the task of final edit generation given an unedited image and its partially edited version obtained by warping. Figure 2 presents an overview of our approach. Since the overall pipeline only requires one forward pass and utilizes efficient foundation models, it takes only a few seconds to edit a pair of images consistently.

Having a method that can edit two images consistently, we extend the approach to editing sets of images by repeating the process for all the remaining query images. We can then pass the resulting set of consistently edited images through a feedforward method such as AnySplat [26] to obtain a 3D Gaussian Splatting (3DGS) representation of the edited scene.

Edited scene geometry estimation We first pass A_{src} and Q_{src} through the reconstruction model Depth Anything 3 [33] to obtain camera intrinsics, K_A and K_Q, respectively, and extrinsics, P_A and P_Q^0. The pose P_Q^0 is then refined into P_Q with the classic relative pose estimation from two views using RoMa [18] matches. The refinement affects the quality of the subsequent joint

Fig. 4: Qualitative comparison of query-anchor consistency with Qwen anchor backbone. Our method GeM-NR preserves photometric and geometric details of the anchor edit.

reconstruction discussed below. Further, we show in Section 4.4 that it improves the quality of the final edits.

Next, the three images $\{A_{src}, A_{edited}, Q_{src}\}$ and the corresponding camera intrinsics and extrinsics $\{(K_A, P_A), (K_{EA}, P_{EA}), (K_Q, P_Q)\}$ are passed through Depth Anything 3, constraining the output cameras to be equal to the ones provided. We use the same camera parameters for A_{edited} as for A_{src}: $K_{EA} = K_A$, $P_{EA} = P_A$. This is based on the assumption that the edit does not change the global motion. E.g., if the edit is: "Change the view to the other side of the corridor" or "Turn to the right and show what is there", it violates our assumption. However, if the edit is: "Turn [depicted object] around", it aligns well with our assumption as it keeps the global scene pose unchanged. See also an example with the edit of the similar form, namely "Have the wooden dinosaur stand next to the stump", in Figure 1. Depth Anything 3 is trained to tackle dynamic scene reconstruction, and we leverage this ability for challenging nonrigid edits. The model views $\{A_{src}, A_{edited}, Q_{src}\}$ as images of a single dynamic scene, where possible changes in geometry and photometry over time appear at A_{edited}.

Edit initialization with warping A partially-filled edited image is rendered at a query viewpoint by projecting the point cloud of the edited scene, which can also be seen as warping A_{edited}, giving Q_{warp}. The warping provides dense guidance on roughly how the edited views should be to ensure consistency with the edited anchor view.

In this work, we use depth map / point cloud representation for both estimation and rendering. In theory, any other representation, such as NeRFs or 3D Gaussians, could work as long as such a representation can be faithfully obtained from only a few images, in our case as few as three images, and the underlying estimation method supports dynamic scenes. Another option is to upgrade to a desired representation starting from a point cloud. A challenge with this approach is that it has access to very limited data, i.e., a single edited image with the corresponding depth map. At this point, we found that manipulating point clouds directly works best.

Why not use monocular depth? A slightly simpler approach could be to use monocular depths estimated from an edited anchor image A_{edited} to warp this image from camera (K_A, P_A) to camera (K_Q, P_Q). In Figure 3, we show qualitative examples of obtaining a monocular depth map from a calibrated edited image as compared to our alignment approach. Ideally, the depth values of A_{src} and A_{edited} should coincide in regions where the geometry has not been changed. In the examples shown, we zoom in on such regions (leaves in the left example and wall in the right example). Monocular depth estimates (even when calibrated) do not follow the multi-view depths of the unedited scene as closely as our approach, in which the depth maps of all images are estimated jointly.

Edit refinement with multi-reference editing To fill in the gaps in Q_{warp}, one could use an image inpainting model. The problem with this approach is that it does not guarantee the consistency of the filled-in areas with those in Q_{src}. We need to be able to constrain the image generation process both by forcing the edit to closely follow Q_{warp} where possible, and also by letting the model infer how to fill the remaining areas based on the semantics of the corresponding regions in Q_{src} and the overall content in Q_{warp}. Hence, the model should be able to accept multiple visual inputs and understand the relation between them from the text description, which we refer to as multi-reference image editing.

Recent image editing methods [6,7,61] allow for multi-reference inputs: a list of images $I_{list} = [I_1, I_2, ...]$ and a text prompt $\hat{T}$ specifying how the images should be combined to create a final image $I_{edited} = f_{multi}\left(I_{list}, \hat{T}\right)$. We propose to include the warping of the initial edited view into the target view as conditioning when editing. This constraint is provided directly in the image frame of the view that should be edited, making it easier for the model to create an image that respects the warping from the initial edited view, encouraging consistency, while also respecting the additional information given in the unedited target view:

$$Q_{edited} = f_{multi}\left([Q_{src}, Q_{warp}], \mathrm{concat}(T, T_+)\right), \tag{1}$$

Fig. 5: Application of our method in interior design.

"Remove the clutter and add a large stylish brown leather armchair standing in the corner but a bit out and facing the room center"

where our additional instruction T_+ is: "The suggested appearance is in the second image. Stick to this change, but refine it to keep consistency with respect to the first image."

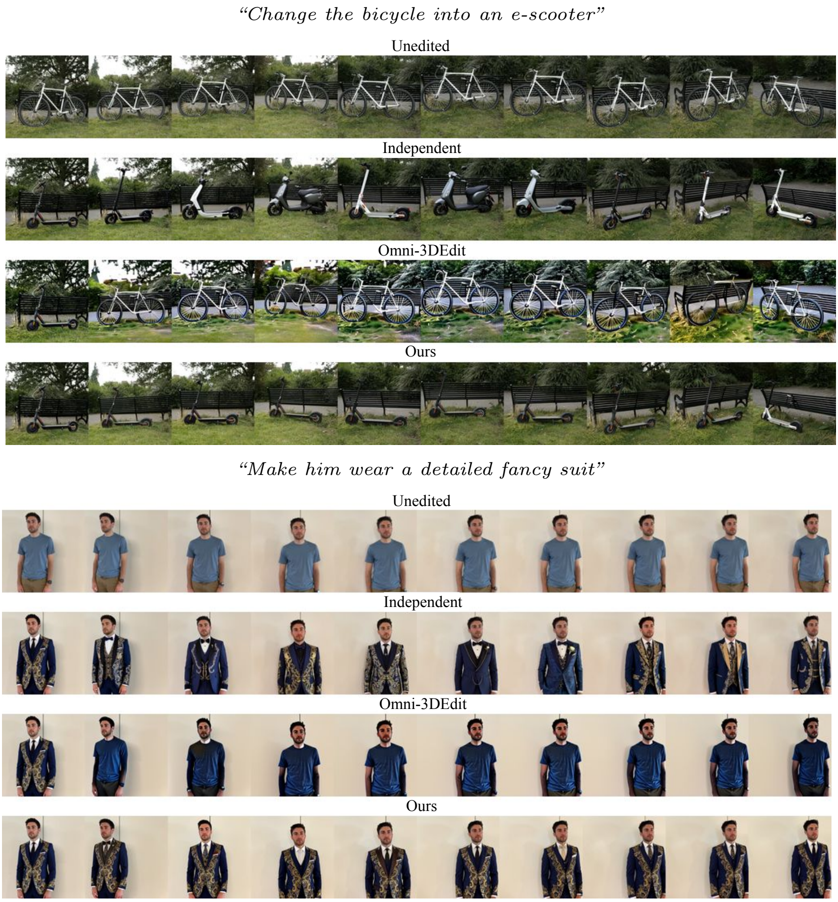
4 Experiments

We divide our evaluation into two main tasks: general multi-view editing (Figures 5-7) and image pair editing (Figure 9). For general multi-view editing, we also evaluate edited 3DGS representations (Figure 7). We evaluate view consistency for both tasks. Additional experimental results can be found in Section A in the supplementary material.

4.1 Experimental setup

We evaluate and compare GeM-NR across a variety of different edit types, on a single NVIDIA A100 with 80GB. Our evaluation data includes 38 prompts over 4 editing categories: general nonrigid edits, object addition, object removal, and appearance change. We use a combination of 15 test scenes taken from the datasets SPIn-NeRF [41], IN2N [22], Mip-NeRF 360 [3], and BlendedMVS [67]. We also report ablation study results on 4 held-out validation scenes, using 11 prompts. For this task, we compare with Omni-3DEdit [37] which performs multi-view editing by directly editing a set of up to 10 images given one edited anchor image. They achieve this by creating a dataset of paired edits for specific scenarios such as object removal, insertion, and appearance change. They can also handle more complex edits by first performing object removal and then object addition. Existing training-free 3DGS-based methods [12, 13, 31] require dense scene capture, limiting direct comparison on the task of sparse-view editing. Sparse-view editing methods based on consistent inpainting, such as MVInpainter [9], do not support general nonrigid editing, so we compare GeM-NR with MVInpainter specifically on object removal, a task it does support. MVInpainter also supports object addition, but this requires manual 3D bounding box annotations, which we do not assume are available.

Fig. 6: Multi-view editing of the 10 image-sequences with Qwen anchor backbone. GeM-NR successfully edits the full sequence while maintaining multi-view consistency.



For evaluating consistent image pair editing, we use image pairs taken from DreamBooth [51] and Mip-NeRF 360 [3], using a total of 38 pairs in the test set. For this task, we compare with Edicho [2], which performs consistent image editing by computing explicit correspondences from the unedited images that are used to guide the denoising process. They apply this approach to two different 2D editing methods, ControlNet [72, 74] for global editing and BrushNet [27] for inpainting-based editing. Their code release only includes the inpainting-based approach [27], so this is the one that we compare to.

Implementation details. We use FLUX.2 [klein] [7] as our backbone multi-reference image editing model. For geometry estimation, we use Depth Anything 3 [33]. For the general multi-view editing, initial anchor edits are generated using Qwen [61]

Fig. 7: Renders from 3DGS of edited scene. Our approach gives sharp renders that preserve the details from the anchor edit.

(version Qwen-Image-Edit-2509) which is the editing method used when training Omni-3DEdit. When comparing image pair editing performance to Edicho, we use BrushNet [27] for the initial anchor edits, since this is the method used by Edicho. BrushNet requires masks and inpaints the masked object area. We adapt our method to work with this type of input by providing a masked anchor edit as a reference, and estimating the object geometry instead of the full scene. Finally, for all tasks, we also test FLUX.2 [klein] [7] for the anchor edits. For generating a 3DGS representation from the edited multi-view images, we use AnySplat [26], a feedforward method that in seconds can generate a 3DGS from a set of sparse unposed images.

GeM-NR offers competitive runtime performance, taking on average ~ 3.4 **seconds per image**, making it comparable to Edicho (3.5 seconds) for pair editing. For multi-view editing, GeM-NR is substantially faster than Omni-3DEdit, which takes ~ 8.3 seconds per image, and the difference is even more pronounced for complex nonrigid edits, where Omni-3DEdit requires ~ 16.6 seconds per im-

Table 1: Consistency evaluation of edited multi-view images. GeM-NR improves multi-view consistency and preserves edit instructions, cf. high TA scores.

		Geometric Edit Consistency		3D Reconstruction Consistency			Text Alignment
		MEt3R↓	mAA, % ↑	PSNR↑	SSIM↑	LPIPS↓	TA / TA dir ↑
All Edit Types ↴							
Qwen	Independent	0.248	31.47	20.70	0.677	0.213	**0.258** / **0.202**
	Omni-3DEdit	0.236	33.99	20.06	**0.690**	0.164	0.230 / 0.127
	GeM-NR	0.194	36.71	21.63	0.653	0.167	0.252 / 0.198
FLUX.2	Independent	0.231	34.03	21.12	0.670	0.203	0.254 / 0.190
	Omni-3DEdit	0.238	34.27	19.57	0.677	0.171	0.233 / 0.122
	GeM-NR	**0.190**	**38.58**	**21.96**	0.659	**0.162**	0.254 / 0.193
	Unedited	0.186	47.71	25.76	0.769	0.116	0.204 / -
General Nonrigid ↴							
Qwen	Independent	0.303	23.15	18.62	0.613	0.284	0.265 / 0.240
	Omni-3DEdit	0.288	23.78	18.21	**0.662**	0.205	0.251 / 0.184
	GeM-NR	**0.215**	31.56	20.68	0.612	0.194	0.268 / **0.252**
FLUX.2	Independent	0.291	24.91	18.93	0.598	0.275	**0.269** / 0.234
	Omni-3DEdit	0.288	21.56	17.60	0.628	0.216	0.248 / 0.172
	GeM-NR	0.216	**32.35**	**20.91**	0.607	**0.188**	0.268 / 0.247
	Unedited	0.216	47.31	25.23	0.745	0.128	0.202 / -
Object Addition ↴							
Qwen	Independent	0.217	41.22	20.72	0.738	0.220	**0.242** / **0.190**
	Omni-3DEdit	0.125	46.60	23.46	**0.780**	**0.100**	0.221 / 0.120
	GeM-NR	**0.120**	46.61	24.40	0.766	0.105	0.226 / 0.151
FLUX.2	Independent	0.165	45.97	22.82	0.769	0.158	0.240 / 0.165
	Omni-3DEdit	0.134	46.99	22.92	0.779	0.107	0.232 / 0.121
	GeM-NR	0.121	**47.17**	**24.78**	0.777	0.111	0.238 / 0.144
	Unedited	0.097	53.33	29.03	0.860	0.070	0.191 / -
Object Removal ↴							
Qwen	Independent	0.166	41.79	24.06	0.734	0.143	0.207 / **0.156**
	MVInpainter	0.177	43.61	**24.85**	**0.751**	0.134	**0.208** / 0.143
	Omni-3DEdit	0.203	38.27	20.83	0.690	0.174	0.201 / 0.114
	GeM-NR	0.153	45.66	24.04	0.712	0.135	0.201 / 0.155
FLUX.2	Independent	0.165	44.62	24.66	0.741	0.142	0.204 / **0.156**
	Omni-3DEdit	0.202	40.90	20.81	0.698	0.170	0.199 / 0.104
	GeM-NR	**0.151**	**46.43**	24.44	0.719	**0.133**	0.204 / 0.154
	Unedited	0.152	48.15	25.95	0.790	0.101	0.203 / -
Appearance Change ↴							
Qwen	Independent	0.256	29.15	**20.74**	0.673	0.193	**0.278** / **0.200**
	Omni-3DEdit	0.257	34.03	19.67	**0.674**	**0.158**	0.232 / 0.099
	GeM-NR	0.224	32.87	20.29	0.614	0.187	0.271 / 0.199
FLUX.2	Independent	0.244	31.27	20.52	0.652	0.199	0.270 / 0.185
	Omni-3DEdit	0.260	34.95	19.09	0.663	0.168	0.237 / 0.098
	GeM-NR	**0.214**	**36.31**	20.63	0.626	0.174	0.270 / 0.193
	Unedited	0.214	46.17	24.80	0.741	0.132	0.210 / -

Fig. 8: 3D reconstructions of the edited images, obtained using VGGT [57].

age. This is because Omni-3DEdit requires two forward passes for nonrigid edits other than object insertion and removal.

Evaluation metrics. For consistent image pair editing, we follow [2] and use CLIP [45] to evaluate text alignment and edit consistency. For general multi-view editing, we evaluate both consistency and text alignment across the edited views. Multi-view consistency is measured with MEt3R [1], which compares DINO [10] embeddings at matches obtained by Dust3R [59], and by 3D reconstruction consistency metrics that measure how well a 3DGS can be fit to the edited images, using PSNR/SSIM/LPIPS to compare against AnySplat [26] renders. To verify that the edited images preserve the same relative poses as for the unedited images, we also report mean average accuracies (mAA) computed by thresholding symmetrized epipolar distances and matching confidences. Finally, we evaluate the resulting 3D Gaussians by measuring how well their renderings align with the text prompt.

4.2 General multi-view editing

Multi-view consistency. We evaluate multi-view consistency when editing a sequence of 10 images. Consistency metrics are reported in Table 1, both overall and per edit category. In general, our method achieves the highest consistency, with the largest improvements for the general nonrigid edits involving significant geometry changes. Omni-3DEdit is limited in scope and struggles with performing the edit for appearance change and general nonrigid edits, as seen by the text alignment scores that measure how well the edit instruction is preserved after multi-view editing. MVInpainter is limited to tasks such as object removal

Table 2: Text alignment (TA↑ / TA dir↑) of the renders from 3DGS of edited scene with Qwen anchor backbone. GeM-NR improves over Omni-3DEdit. It especially improves for general nonrigid edits and object addition, where independent and Omni-3DEdit struggle. Comparable results are achieved when using FLUX.2 [klein] anchor backbone, as can be seen in Section A.5 in the supplementary material.

	All Edit Types	General Nonrigid	Object Addition	Object Removal	Appearance Change
Independent	0.250 / 0.191	0.249 / 0.214	0.225 / 0.134	**0.212** / 0.170	**0.275** / **0.206**
Omni-3DEdit	0.229 / 0.130	0.244 / 0.194	0.232 / 0.130	0.207 / 0.126	0.226 / 0.092
MVInpainter	-	-	-	0.209 / 0.132	-
GeM-NR	**0.258** / **0.212**	**0.269** / **0.273**	**0.246** / **0.188**	0.211 / **0.173**	0.274 / 0.198

Table 3: Consistent image pair editing (inpainting) evaluation.

		mAA, % ↑	MEt3R (object) ↓	TA / TA dir (object) ↑	EC (object) ↑
DreamBooth images					
BrushNet (uses masks)	Independent	-	0.646	0.279 / 0.218	0.834
	Edicho	-	0.324	**0.287** / 0.244	0.887
	GeM-NR	-	0.258	0.280 / **0.245**	0.899
FLUX.2 (with masks)	Independent	-	0.528	0.271 / 0.210	0.871
	GeM-NR	-	**0.244**	0.268 / 0.224	**0.902**
Mip-NeRF 360 test scenes					
BrushNet (uses masks)	Independent	23.68	0.500	**0.252** / 0.177	0.885
	Edicho	28.13	0.319	0.248 / 0.178	**0.941**
	GeM-NR	28.04	0.285	0.248 / 0.179	0.920
FLUX.2 (with masks)	Independent	30.22	0.271	0.242 / 0.193	0.905
	GeM-NR	**33.89**	**0.217**	0.238 / **0.198**	0.920

that can be formulated as inpainting. In contrast, our method successfully handles a broad range of edits. Figure 6 shows that GeM-NR is able to successfully perform the multi-view edits, while Omni-3DEdit fails at performing the edit and independent edits lead to significant inconsistencies. The top two rows of Figure 4 show examples where Omni-3DEdit successfully performs the edit, but GeM-NR better preserves consistent details and appearance with the anchor edit. For some edit types, e.g., appearance change, Omni-3DEdit gets better SSIM and LPIPS scores than GeM-NR. A reason for this is that these metrics can favor weaker edits that are inherently more consistent, as can be seen in our method getting higher TA scores. In Figure 5 we also show an example of how our method can perform consistent multi-view editing for use-cases in interior design.

Edited 3D representations. The edited multi-views can also be used to generate a 3D Gaussian representing the edited scene. Table 2 shows that GeM-NR gives edited 3D Gaussians that best align with the desired edit instruction, with the largest improvements for object addition and general nonrigid edits where there are significant changes in geometry. In Figure 7 we observe that GeM-NR gives renders that are sharp and clearly preserve the details in the edited anchor view.

Fig. 9: Qualitative image pair editing examples for Edicho and our GeM-NR.

The renderings from these 3D Gaussians can be found on the project page. We also show in Figure 8 that a 3D reconstruction of the edited scene can be obtained by using VGGT [57] on the edited images.

4.3 Image pair editing

Table 3 shows evaluation results for image pair editing on DreamBooth images and Mip-NeRF 360 test scenes. MEt3R, TA, and EC are computed on the masked images where only the object is kept (since the background varies a lot between the images, see top row in Figure 9). GeM-NR achieves comparable or better consistency as well as text alignment.

4.4 Ablation Studies

Table B.1 shows the ablation configurations and results on the held-out validation dataset. We evaluate several modifications to the proposed method: (1) Reference text (Input edit): whether to provide the original edit description; if not, provide "Edit the first image in the same way as shown in the second image"; (2) Reference image(s): which of the images to provide: original query Q_{src}, edited anchor A_{edited}, and depth-warp Q_{warp}. We also compare our method to the baseline approach (first row) that concatenates the two inputs into one image and asks the model to edit the two images consistently. Finally, we evaluate GeM-NR without pose refinement (last row). GeM-NR is highlighted in gray. It achieves a trade-off across all performance evaluation metrics, as indicated in the last column that reports a balanced score: $\texttt{Balanced} = \texttt{mAA} \cdot (1 - \texttt{MEt3R}/2) \cdot \texttt{TA} \cdot \texttt{EC}$.

Table 4: Ablation study of consistent image pair editing. Our method is highlighted in gray. It achieves a trade-off across all performance evaluation axes.

Reference text	Reference image(s)			Metrics				
Input edit	Original query	Edited anchor	Depth-warp	MEt3R ↓	mAA, % ↑	TA ↑	EC ↑	Balanced ↑
Simultaneous pair editing								
Concatenate both inputs								
✓	N/A	N/A	N/A	0.229	35.40	0.271	0.888	8.08
Editing one conditioned on another								
Original edit text prompt preserved; no warp								
✓	✓	✗	✗	0.238	31.70	0.214	0.924	7.43
✓	✓	✓	✗	0.069	13.53	0.271	0.888	3.53
Original edit text prompt ignored; with warp								
✗	✗	✗	✓	0.219	39.83	0.268	0.827	8.47
✗	✓	✗	✓	0.220	40.74	0.268	0.844	8.87
✗	✗	✓	✓	0.231	38.74	0.266	0.823	8.25
✗	✓	✓	✓	0.185	26.11	0.271	0.863	5.92
Original edit text prompt preserved; with warp								
✓	✓	✓	✓	0.173	23.96	0.275	0.913	5.86
✓	✓	✗	✓	0.199	40.63	0.274	0.885	**9.50**
No relative pose refinement				0.158	39.88	0.278	0.894	9.28
Unedited				0.190	49.72	0.214	0.924	-

The detailed instructions used for the different methods can be found in Section B in the supplementary material.

5 Conclusion

We present GeM-NR, a method for multi-view consistent edits that can substantially change scene geometry and appearance. Our approach is flexible, handling diverse editing tasks, and fast, requiring only a few seconds per image. Extensive evaluations on different edit categories show that GeM-NR improves multi-view consistency and that the edited images can be used to generate 3D representations that align well with the specified edit instruction.

Our current approach conditions multi-view editing on a single anchor edit, which does not explicitly enforce consistency between query views. While this strategy works well for scenes with limited viewpoint variation, it becomes less effective for larger scenes with more extreme viewpoint changes. A potential solution is to condition the editing not only on one anchor view, but also on other previously edited images.

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Supplementary Material

This supplementary material presents additional experimental results (Section A) and detailed prompts used by each method in the ablation study (Section B). Qualitative video results can be found on the project page: https://gem-nr.github.io.

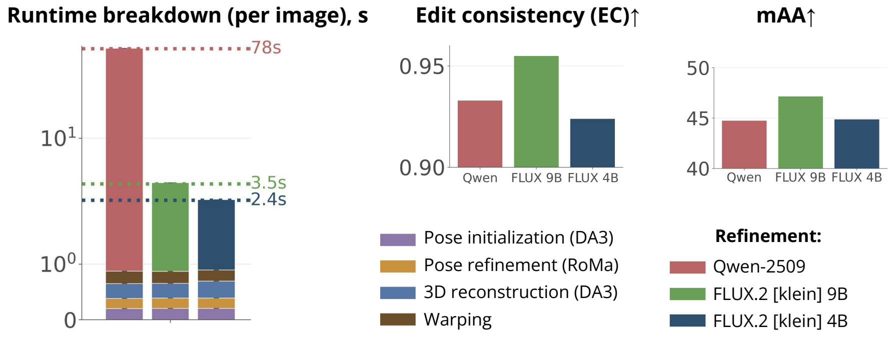
A Additional Experimental Results

Below, we provide additional experimental results, including evaluation of the 2D editing backbones; analysis of sensitivity to increased disparity, reconstruction errors, and global motion change; and further qualitative examples of edited images and renders.

A.1 Backbone choice

There are two types of 2D editing backbones that can be utilized within the GeM-NR framework: one for *anchor editing* and one for *refinement after warping*. For anchor editing, any 2D editor can be used, and in the main paper, we evaluate multiple state-of-the-art models. For refinement after warping, the 2D editor

Fig. A.1: Runtime breakdown and consistency evaluation on object addition task. Runtime is shown on a semi-logarithmic scale, excluding anchor editing. The reported consistency metrics are CLIP-based (EC) and epipolar distance-based (mAA). FLUX.2 [klein] 9B offers a good trade-off between runtime and performance.



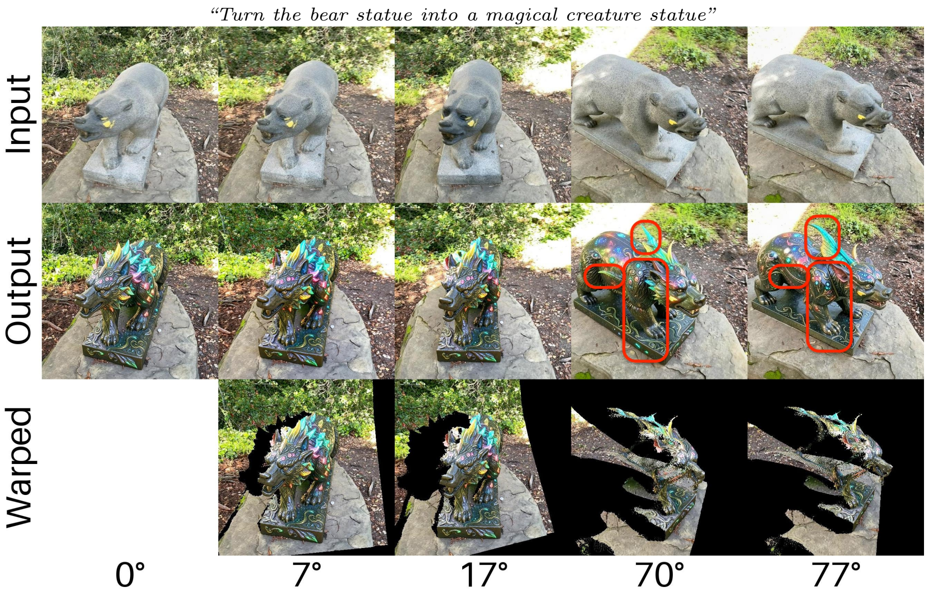
must support multi-reference. We limit our choice to FLUX.2 or Qwen models, which are recent and commonly used methods for this task. We evaluate these models on the object addition task and provide a breakdown of their runtime and performance in Fig. A.1. FLUX.2 9B offers a good trade-off between speed and performance, motivating our default choice of FLUX.2 9B as the refiner.

We find that GeM-NR performs best when the anchor and refinement editors match (see Table 1). This is unsurprising: the refiner is more likely to produce an image consistent with the edited anchor when it can reproduce the same edit as the anchor editor.

A.2 Sensitivity to increased disparity

Fig. A.2 illustrates an example of editing a query image with an increasingly wide baseline relative to the anchor image. This results in larger missing regions in the warped images, providing less reference information for refinement and greater freedom for hallucination. Consequently, query images forming a wide baseline with the anchor image have a higher risk of localized inconsistencies, as illustrated in Fig. A.2 (70°–77°). Addressing this is a promising future research direction.

Fig. A.2: Effects of increased disparity on warped/edited images.



A.3 Sensitivity to reconstruction errors

Fig. A.3 shows an example in which Depth Anything 3 fails to accurately reconstruct the scene, resulting in flatter facial surfaces. We find that reconstruction errors frequently occur in facial regions. GeM-NR is designed to be modular and can therefore readily accommodate future upgrades to the underlying backbones.

However, we also observe that GeM-NR in its current form is often robust to such errors (see Fig. A.3). When the warped region is implausible, FLUX tends to rely less on it and more on the text prompt and input image.

A.4 Sensitivity to global motion change

While our problem setting naturally assumes unchanged global motion, we tried to violate this assumption in order to assess the sensitivity of GeM-NR to global motion change. We simulate this scenario by instructing the model to move the camera, as shown in Fig. A.4. The resulting edit is visually pleasing but exhibits a small, barely noticeable error, stemming from an expected misalignment between the original and edited point clouds (Fig. A.4, bottom left). A straightforward fix is to incorporate A. ↔ E.A. relative pose estimation, which ensures improved alignment (see Fig. A.4, bottom right) and greater 3D consistency during editing.

Fig. A.3: DA3 error example giving flatter face surface. FLUX ignores this error.

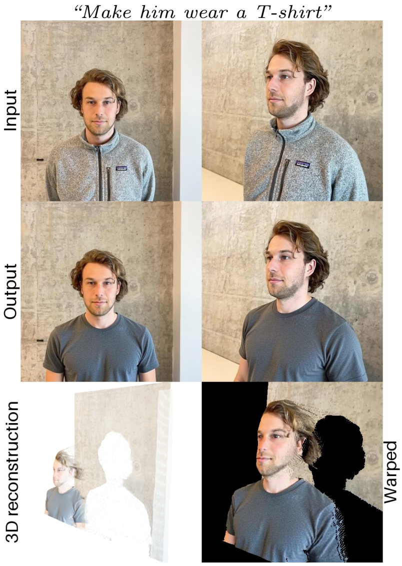


Fig. A.4: Global motion change: results without and with global motion estimation.

A.5 Additional evaluation results

In Table A.1, we report text alignment metrics for renderings of the edited 3DGS when FLUX.2 [klein] is used to edit the initial anchor image. The results are comparable to those obtained when using Qwen to edit the anchor image (see

Table 3). Additional qualitative examples of the multi-view consistent editing are provided in Figures A.5, A.6, A.7.

Table A.1: Text alignment (TA↑ / TA dir↑) of the renders from 3DGS of edited scene with FLUX.2 [klein] anchor backbone. GeM-NR clearly improves over Omni-3DEdit. It especially improves for the general nonrigid editis and object addition, where both Independent and Omni-3DEdit struggle.

	All Types	General Nonrigid	Object Addition	Object Removal	Appearance Change
Unedited	0.201 / -	0.194 / -	0.201 / -	0.200 / -	0.206 / -
Independent	0.253 / 0.192	0.254 / 0.227	0.243 / 0.169	**0.213** / **0.170**	0.271 / 0.187
Omni-3DEdit	0.233 / 0.131	0.244 / 0.185	0.240 / 0.140	0.206 / 0.114	0.233 / 0.101
Ours	**0.259** / **0.205**	**0.270** / **0.269**	**0.253** / **0.176**	0.211 / 0.163	**0.272** / **0.190**

Fig. A.5: Edited multi-view images. GeM-NR gives consistent edits with preserved details, such as the bicycle parts and the patches on the jacket.

Fig. A.6: Additional qualitative multi-view editing results with Qwen anchor editor.

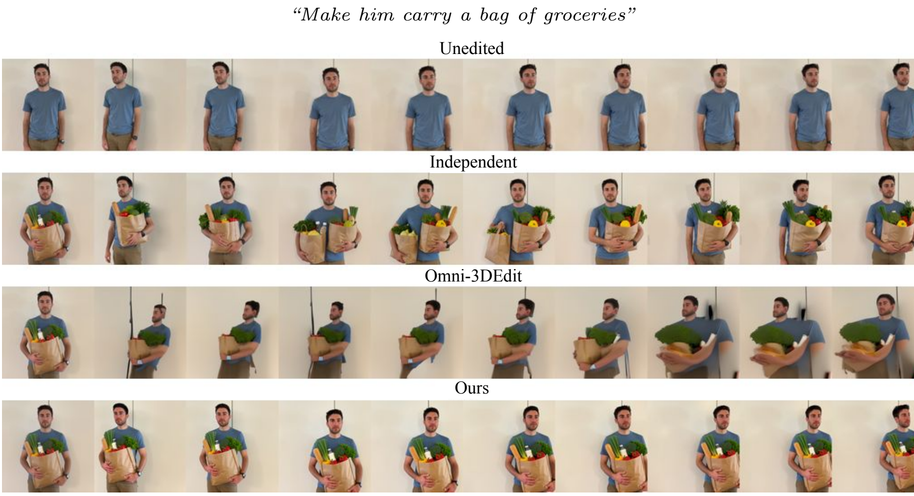
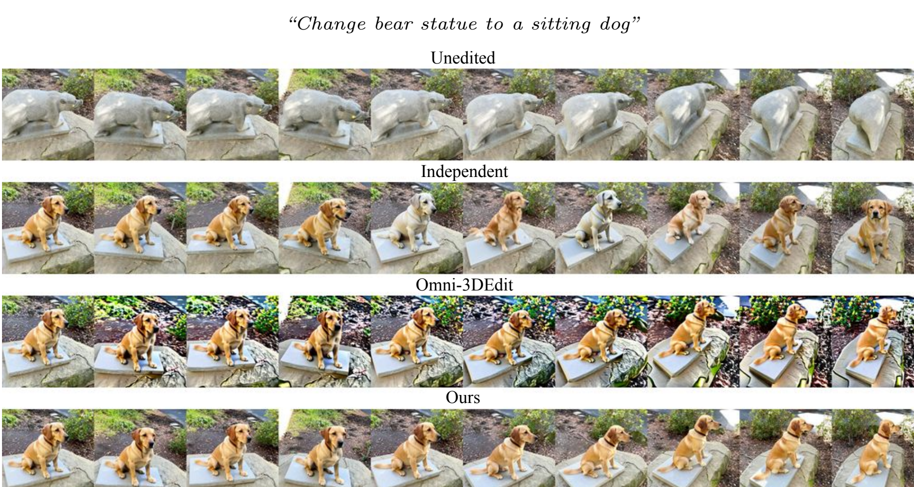


Fig. A.7: Additional qualitative multi-view editing results with Qwen anchor editor.



B Ablation Details

Table B.1 presents the detailed instructions used by each method in the ablation study. These instructions are used during the refinement stage. GeM-NR is highlighted in gray.

Table B.1: Instructions corresponding to each method in the ablation study.

Reference text	Reference image(s)			Instruction
Input edit	Original query	Edited anchor	Depth-warp	
Simultaneous pair editing				
Concatenate both inputs				
✓	N/A	N/A	N/A	`“Edit these two concatenated images in a consistent way”`
Editing one conditioned on another				
Original edit text prompt preserved; no warp				
✓	✓	✗	✗	`T`
✓	✓	✓	✗	`“Edit the first image as shown in the second image”`
Original edit text prompt ignored; with warp				
✗	✗	✗	✓	`“Inpaint this image”`
✗	✓	✗	✓	`“Edit this image as shown in the second image. Inpaint the second image”`
✗	✗	✓	✓	`“Inpaint the first image. Use the second image as a guidance on how to inpaint”`
✗	✓	✓	✓	`“Edit the first image in the same way as shown in the second image. The suggested appearance is in the third image. Stick to this change, but refine it to keep consistency with respect to the first image”`
Original edit text prompt preserved; with warp				
✓	✓	✓	✓	`concat(T, “The suggested appearance is in the second image. Stick to this change, but refine it to keep consistency with respect to the first image. For reference, the same edit at a different viewpoint is provided in the third image”)`
✓	✓	✗	✓	`concat(T, “The suggested appearance is in the second image. Stick to this change, but refine it to keep consistency with respect to the first image”)`